

CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Lecture 4

1 Simple Linear Regression: Normal Homoskedastic Error Case

- Assumptions
- Least Squares
- Properties of Ordinary Least Squares (OLS) Estimators
 - Unbiasedness OLS Estimators
 - Variances and Covariance of OLS Estimators
 - Confidence Intervals and Hypothesis Testing
- Measure of Fit
 - ESS, TSS, and R Squared
 - Adjusted R Squared

2 Econometric Models: Linear Regression

- Terminology and Assumptions
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Large Sample Distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$

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Simple Linear Regression: Normal Homoskedastic Error

Assumptions:

For $i \in \{1, 2, \dots, n\}$, let

- x_i be fixed values with $\min\{x_i\} < \max\{x_i\}$
- $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, where σ^2 is independent of i (*homoskedasticity*)
- $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

Simple Linear Regression: Normal Homoskedastic Error

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- $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

Note:

Given x_i ,

$y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$ are independent, where $i \in \{1, 2, \dots, n\}$.

Then

- One can use, for example, MLE to estimate β_0 and β_1 .
- Given *predictor* x_i , the expectation of *response* y_i is

$$E[y_i] = \beta_0 + \beta_1 x_i.$$

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Simple Linear Regression: Least Squares

Claim:

Given a Simple Linear Regression model,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i \in \{1, 2, \dots, n\},$$

the Maximum Likelihood Estimators (MLE's) of β_0 and β_1 can be equivalently obtained by minimizing the sum of squared residuals,

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \underbrace{[\beta_0 + \beta_1 x_i]}_{E[y_i]})^2,$$

and are

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Note:

These estimators are often referred as *ordinary least squares* (OLS) estimators.

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Properties of OLS Estimators: Unbiasedness

Thm.

Given a Simple Linear Regression model,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i \in \{1, 2, \dots, n\},$$

the ordinary least squares (OLS) estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased:

$$E[\hat{\beta}_0] = \beta_0 \quad \text{and} \quad E[\hat{\beta}_1] = \beta_1.$$

Properties of OLS Estimators: Variances and Covariance

Thm.

Given a Simple Linear Regression model,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i \in \{1, 2, \dots, n\},$$

for the ordinary least squares (OLS) estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ we have:

$$\text{Var}(\hat{\beta}_0) = \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = - \frac{\sigma^2 \sum_{i=1}^n x_i}{n \sum_{i=1}^n (x_i - \bar{x})^2}$$

Properties of OLS Estimators: CIs and Hypothesis Testing

Def.

Given a Simple Linear Regression model,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i \in \{1, 2, \dots, n\},$$

and the ordinary least squares (OLS) estimators $\hat{\beta}_0$ and $\hat{\beta}_1$, we introduce

- *residual sum of squares* (RSS):

$$\text{RSS} \doteq S(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^n (y_i - \underbrace{[\hat{\beta}_0 + \hat{\beta}_1 x_i]}_{\hat{y}_i})^2$$

- estimator of σ^2 :

$$s^2 = \frac{\text{RSS}}{n - 2}$$

- estimators of $\text{Var}(\hat{\beta}_1)$ and $\text{Var}(\hat{\beta}_0)$:

$$s_{\hat{\beta}_0}^2 = \frac{s^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and} \quad s_{\hat{\beta}_1}^2 = \frac{s^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Properties of OLS Estimators: CIs and Hypothesis Testing

Claim:

Given a Simple Linear Regression model,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i \in \{1, 2, \dots, n\},$$

we have

$$\frac{\hat{\beta}_0 - \beta_0}{s_{\hat{\beta}_0}} \sim t_{n-2} \quad \text{and} \quad \frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}} \sim t_{n-2}.$$

Properties of OLS Estimators: CIs and Hypothesis Testing

Corollary:

$(1 - \alpha)100\%$ confidence interval for β_k with $k = 0, 1$ is:

$$\beta_k = \hat{\beta}_k \pm t_{n-2, \alpha/2} s_{\hat{\beta}_k}.$$

Here, $t_{n-2, \alpha/2}$ denotes a number such that $F_{t_{n-2}}(t_{n-2, \alpha/2}) = 1 - \alpha/2$, where $F_{t_{n-2}}(\cdot)$ denotes the cdf of the t_{n-2} distribution.

Properties of OLS Estimators: CIs and Hypothesis Testing

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Corollary:

To test $H_0 : \beta_k = 0$ vs. $H_1 : \beta_k \neq 0$, one can look at the following test statistic:

$$TS \doteq \frac{\hat{\beta}_k - 0}{s_{\hat{\beta}_k}},$$

which follows t_{n-2} under $H_0 : \beta_k = 0$.

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Measure of Fit: ESS, TSS, and R^2

Def.

Let $\hat{\beta}_0$ and $\hat{\beta}_1$ denote the OLS estimators and let

$$\hat{y}_i \doteq \hat{\beta}_0 + \hat{\beta}_1 x_i \quad \text{for all } i \in \{1, 2, \dots, n\},$$

then we define

- *explained sum of squares* (ESS):

$$\text{ESS} \doteq \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

- *total sum of squares* (TSS):

$$\text{TSS} \doteq \sum_{i=1}^n (y_i - \bar{y})^2$$

- coefficient of determination:

$$R^2 = \frac{\text{ESS}}{\text{TSS}}$$

Measure of Fit: ESS, TSS, and R^2

Claim:

The coefficient of determination

$$R^2 = 1 - \frac{SSR}{TSS},$$

where

$$SSR \doteq \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

is the sum of squared residuals.

Note: $TSS = ESS + SSR$.

Measure of Fit: R^2_{adj}

When a variable is added to the model, R^2 increases, which makes it hard to use R^2 in selecting the optimal number of predictors. A modification to the coefficient of determination is introduced as follows:

Def.

Adjusted R^2 is defined as

$$R^2_{\text{adj}} \doteq 1 - \frac{s^2_{\hat{\varepsilon}}}{s^2_y},$$

where

$$s^2_{\hat{\varepsilon}} \doteq \frac{1}{n - k - 1} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \text{and} \quad s^2_y \doteq \frac{1}{n - 1} \sum_{i=1}^n (y_i - \bar{y})^2.$$

Here, k denotes the number of variables of the model:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots + \beta_k x_{k,i} + \varepsilon_i.$$

Measure of Fit: R^2_{adj}

Claim:

Given a model

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots \beta_k x_{k,i} + \varepsilon_i,$$

the *adjusted* R^2 can be computed as follows:

$$R^2_{\text{adj}} = 1 - \frac{SSR/(n - k - 1)}{TSS/(n - 1)}.$$

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Linear Regression: Terminology and Assumptions

Terminology:

We assume now that both X_i and Y_i are random and there is relation

$$Y_i = \beta_0 + \beta_1 X_i + u_i,$$

where

- index $i \in \{1, 2, \dots, n\}$
- Y_i is called *dependent variable* or *response*
- X_i is called *independent variable*, *regressor*, or *explanatory variable*
- function $\beta_0 + \beta_1 x$ is referred as *population regression line*
- β_0 is *intercept* of the population regression line
- β_1 is *slope* of the population regression line
- u_i is the *error term*

Linear Regression: Terminology and Assumptions

Assumptions:

We assume that there is the following relation between X_i and Y_i :

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

where

- 1 *exogeneity* assumption: $E[u_i|X_i] = 0$
- 2 (X_i, Y_i) is i.i.d. sample drawn from a joint distribution
- 3 $E[X_i^4]$ and $E[Y_i^4]$ are nonzero and finite

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OLS Estimators

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

the OLS estimators are given by

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad \text{and} \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X},$$

where both X_i and Y_i are now random.

Unbiasedness of the OLS Estimators

Thm.

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased:

$$E[\hat{\beta}_0] = \beta_0 \quad \text{and} \quad E[\hat{\beta}_1] = \beta_1.$$

Large Sample Distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$

Thm.

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

as $n \rightarrow \infty$, we have:

- $\hat{\beta}_1 \xrightarrow{D} N(\beta_1, \sigma_{\hat{\beta}_1}^2)$, where

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\text{Var}((X_i - \mu_X)u_i)}{(\text{Var}(X_i))^2}$$

- $\hat{\beta}_0 \xrightarrow{D} N(\beta_0, \sigma_{\hat{\beta}_0}^2)$, where

$$\sigma_{\hat{\beta}_0}^2 = \frac{1}{n} \frac{\text{Var}(H_i u_i)}{(\text{E}(H_i^2))^2}, \quad \text{where } H_i = 1 - \left[\frac{\mu_X}{\text{E}[X_i^2]} \right] X_i.$$